System One Decision Models

Availability, modalities, source code, model formats, on-device deployment and research papers

Purpose. A practical research brief for comparing System One-style decision models and assessing whether they can be deployed locally, including on an NPU.

> **How to read this report. “Open weights” does not automatically mean open training code, an official ONNX export, or NPU compatibility. Model availability and format claims should be verified against the exact checkpoint and its license.**

Executive summary

- **Laya** is a useful first conversion experiment because community projects describe ONNX and hardware-specific ports.
- **Kev** is a useful open-weight comparison, but a generic ONNX/NPU build should not be assumed for every variant.
- **Jev** is API-first in the reviewed materials; no public weights were identified for the original model.
- Multimodal models exist, but native audio decision support was not sufficiently verified for the main text-oriented checkpoints.
- A model is deployable on an NPU only when the full graph, preprocessing and post-processing are supported by the target runtime and accelerator.

Evidence caveat: This document formats the findings and links from the preceding research response. Registry entries and newly published preprints can change quickly; treat “unverified” as a request for further confirmation, not proof that a format or capability is impossible.

1. Main models: access and deployment outlook

The core distinction is between hosted APIs, downloadable weights, and independent implementations that reproduce the general typed-decision approach.

Model	Weights / implementation	Modalities reported	On-device outlook
Jev — TypeSafe AI	No public weights identified; hosted API and SDK/API integration	Text/JSON; image support is model-specific at API level	Not directly deployable locally without weights or an authorized implementation.
Laya	Open weights; public Python implementation/package	Text decisions	Strong first candidate; ONNX and several community deployment paths are listed.
Kev	Open weights; public implementation and local serving options	Text decisions	Good local comparison; exact export/NPU compatibility must be checked per variant.
Clef / Clef-flash	Published weights and hosted inference reported	Image + text reported for family	Promising multimodal candidate; verify exact checkpoint and runtime.
Clef-Omni	Exact release artifacts require verification	Do not assume every modality from the family name	Unconfirmed until weights, model card and inference pipeline are checked.
Other open decision models	Varies by project; many publish weights	Mostly text; some image/video research models	Ranges from straightforward local execution to substantial conversion work.

Sources: Jev/API documentation · System One API reference · Laya listing · Kev listing · Open implementations directory.

2. Registry of models

The preceding research response identified an independent registry updated October 22, 2026, with 30 entries from 25 makers. The table below reproduces the listed entries and deployment screening notes; it is not a canonical inventory of every research prototype.

Model	Listed size	Availability / modalities	Local-deployment screening note
Vega	0.8B; project also reports 4B	Open weights; text + image reported	Local candidate; verify exact checkpoint and vision export.
AJEV	26B	Open weights; text/typed decisions; image/audio not established	Possible on capable hardware; difficult edge target.
Nativ-HE-Decision	364M	Open weights; Hebrew-first text decisions	Compact candidate; verify export and language fit.
Strom	Not specified	Hosted API; proprietary; text decision interface	API-first; local path blocked unless weights are released.
Canopy-Jev	27B	Open weights; text	Possible local execution, but large for edge.
Docto-Decision	4.7B	Open weights; text	Check code completeness, export and operators.
Gutsy	800M	Open weights; text	Promising size for local tests; validate quantization.
ModernJev-Decision	150M	Open weights; text	Compact; confirm task coverage and quality.
StartLux-Decision	4.7B	Open weights; text	Local candidate with relatively high memory needs.
Bekko-System-One	395M	Open weights; text	Compact; exact artifact formats need checking.
BioDecision	4B	Open weights; biomedical text decisions	Check domain fit before deployment.
Bev-Decider	478M	Open weights; text	Compact; verify tokenizer and inference code.
Rizzo-Flow	4B	Open weights; text	Check export and runtime compatibility.
Security-One	27B	Open weights + hosted API; text/security decisions	Large; possible self-hosting, challenging edge target.
GLiNER2-5-DECIDE	340M	Open weights + hosted API; text and extraction-related tasks	Compact candidate; verify exact checkpoint.
Decider	1.9B registry entry; other variants exist	Open weights/code; text; separate vision variants reported	Do not assume all variants are multimodal or exportable.
Julia-1	144M	Open weights; text	Attractive compact candidate for initial edge tests.
CLM	8B	Open weights; text	Edge feasibility depends on memory and quantization.
NeoHorse-Jev	4B	Open weights; text	Check checkpoint and runtime compatibility.

Model	Listed size	Availability / modalities	Local-deployment screening note
Lumma-FEV	649M	Open weights; text	Compact; verify artifact availability.
Kev	4B registry entry; family also lists 0.8B, 9B, 27B	Open weights/code; text	Good local comparison; export is variant-specific.
Laya	421M	Open weights/code; text	Strong conversion candidate; community ports are listed.
Open-Jev-DeBERTa-v3-large	434M	Open weights; text	Encoder-style candidate; verify exact checkpoint's ONNX support.
Mira	Not specified	Open weights; text	Local feasibility plausible; artifact details needed.
Caliber-Flex-1B	Not specified	Open weights; text	Potential candidate; format/code completeness unverified.
Caliber-Flex-5B	Not specified	Open weights; text	Higher memory needs; exact format unverified.
Laya Typed-Decisions NoulXP	421M	Open weights; text	Package does not itself guarantee NPU acceleration.
Laya NoulXP-GPU	421M	Open weights; text	GPU-oriented; not automatically suitable for a different NPU.
Laya NoulXP-CPU	421M	Open weights; text	Useful CPU baseline before NPU testing.
Laya Multilingual NoulXP	322M	Open weights; multilingual text	Verify language coverage and export support.

Registry source: System One model registry. Sizes and model claims should be checked against the exact linked model card; reported benchmark results may use different datasets and hardware.

3. Modality support: text, images, video and audio

Text: Jev, Laya and Kev are the primary starting points for text-based routing, classification, scoring and yes/no judgments. Native image/audio support should not be inferred from a text checkpoint's name.

Images: Clef/Clef-flash, Vega and selected research implementations are reported as image-capable. Confirm that the exact checkpoint accepts images and that its vision encoder and preprocessing are included in the local package.

Video: VideoJev and MetaEncoder are research directions for video/multimodal decisions. The cited work describes large models, so they are less immediately suitable for constrained edge devices.

Audio: Native audio-input decision support was not sufficiently verified for the main text-oriented checkpoints. Three different configurations must be distinguished:

- **Native audio model:** waveform/audio features enter the model directly.
- **Audio-to-text pipeline:** a speech-recognition model transcribes audio, then the decision model processes text.
- **Multimodal parent model:** the backbone supports audio, but a specific decision checkpoint or API may not expose it.

The second configuration can work on-device, but adds another model, latency and transcription errors. Verify modality support from the model card and the API's supported-input list.

Useful links

- System One API reference
- Open implementations directory
- System One model registry
- VideoJev paper
- MetaEncoder paper

4. Model formats and runtimes

Artifact / runtime	Meaning	Evidence and deployment relevance
Hugging Face / SafeTensors	Weights plus config/tokenizer assets	Common starting point for open-weight models; not optimized for mobile/NPU by itself.
PyTorch / Python package	Reference execution and preprocessing code	Good correctness baseline; may need export or hardware-specific backend.
ONNX	Interchange graph format	Promising if the full graph exports and the chosen runtime/NPU supports its operators.
Core ML	Apple model format/runtime	Relevant to Apple devices; not a vendor-neutral NPU format.
MLX	Apple Silicon ML framework	Useful for Apple Silicon; not a general Android NPU route.
GGUF	Container commonly used by llama.cpp-family tooling	Can help compatible CPU/GPU execution; NPU support depends on backend.
NPU-compiled artifact	Compiled for a specific vendor SDK/compiler	Direct route to that accelerator but generally not portable to another NPU.
NoulXP / System One package	Model packaging/interoperability conventions	May simplify packaging/serving; does not guarantee NPU graph support.

Format/implementation census: open implementations · curated model/runtime census.

Concrete deployment evidence reported

Laya — ONNX Runtime. A public project describes local execution through ONNX Runtime, including a Go implementation.

Laya — Core ML / Neural Engine. A separate implementation targets Apple's Core ML and Neural Engine; this does not establish portability to other NPUs.

Laya — Axera NPU. A community port for AX8850/NPU3 is listed, with ONNX export and Pulsar2 build steps. It is evidence of a hardware-specific port, not universal NPU support.

WaterSheep — browser ONNX. A ModernBERT-based decision model is described as having an ONNX build for browser execution. Browser execution alone does not prove native NPU acceleration.

Evidence index: awesome-system-one-models and open implementations directory.

5. Research papers to read

These are recent papers/preprints identified in the preceding research response. Read their methods, model versions and evaluation setups carefully; new work in this area is evolving quickly.

1. Evaluating and Benchmarking the System One Model Jev

Tobias Deußer, Lorenz Sparrenberg and Rafet Sifa · September 2026
Evaluates Jev across 37 datasets, compares with Qwen and Gemma baselines, and studies calibration and sensitivity to question changes/removal.
Why read it: Read for baseline quality, probability calibration and failure modes.

2. Typed Decision Models: An Early Evidence Audit and Evaluation Checklist

Lijuan Tang and Yuemeng Zheng · September 2026
Reviews early typed-decision work and proposes an evaluation checklist; cautions that a typed interface alone does not prove an accuracy advantage.
Why read it: Read for fair evaluation design and avoiding unsupported architecture claims.

3. FastJEV: Understanding Redundancy for Compact JEV Inference

October 2026
Studies repeated candidate computation, intermediate-state reuse and pruning to reduce inference work.
Why read it: Read for inference efficiency; it is not primarily an NPU-export paper.

4. MetaEncoder: Exploring the Limit of Bi-Encoders for Multimodal System One Decision Making with Natural Language Interface

October 2026
Explores candidate scoring with image/video inputs and bi-encoder designs for large candidate spaces; described model is large.
Why read it: Read for multimodal architecture and candidate-independent encoding.

5. VideoJev: System One Model for Video Understanding and Interaction

October 2026
Reframes video question answering and temporal grounding as structured decisions; explores visual-computation reuse.
Why read it: Read for multimodal decisions and reuse of visual computation.

6. Releasing the Inner Jev: A Simple Recipe for Open Decision Models

October 2026
Discusses reproducing the typed-decision pattern with open implementations rather than a proprietary endpoint.
Why read it: Read for training/replication approaches; compare implementation and evaluation before claiming equivalence to Jev.

Additional resources

• Awesome Jev Papers — bibliography spanning open models, evaluation, calibration, robustness and multimodal decisions.

• S1MB: System One Mosaic Benchmark — comparison resource; inspect task composition and methodology before interpreting leaderboard results.

6. Recommended on-device investigation

Priority	Candidate	Reason
1	Laya	Best initial conversion experiment due to listed community ONNX and hardware-specific paths. Test accuracy against your task.
2	Compact encoder model (e.g., Julia-1 or Open-Jev-DeBERTa-v3-large)	Useful comparison; encoder-style graphs may be easier to deploy, but operator support and quality must be checked.
3	Kev	Open-weight comparison model; treat NPU export as a separate investigation.
4	Clef / Valen / VideoJev	Investigate if image or video inputs are required. Multimodal export includes the vision pipeline and is a larger task.

Go/no-go checklist

- Obtain exact weights, model configuration, tokenizer, adapters/decision head and license.
- Run the reference implementation locally and save probability outputs for a fixed evaluation set.
- Export to ONNX or another supported format if a suitable artifact does not already exist.
- Compare exported probabilities against reference outputs—not only the winning option.
- Compile for the target NPU and verify it actually runs on the accelerator rather than silently falling back to CPU.
- Measure decision agreement, Brier score/ECE, latency, peak memory, power and stability on realistic inputs.

Bottom line

> **Laya is the clearest first export experiment; Kev is a useful open-weight comparison; and Jev remains API-first based on the evidence reviewed. For image/video decisions, investigate a model with an explicitly multimodal checkpoint. For raw audio, treat native support as unverified unless the model card and inference code prove it.**

Final constraint: the actual target NPU and operating environment determine the viable runtime, supported operators and numerical precision. Hardware-specific validation is required before claiming deployment readiness.

Source index

- System One API reference
- Jev / TypeSafe AI documentation
- Laya model listing
- Kev model listing
- System One model registry
- Open implementations directory
- Curated model/runtime census
- Awesome Jev Papers